

# \* Coefficient of variation

1) 10, 20, 30, 40, 50

Mean = 30  
 Variance = 200  
 Stand Dev = 14.14  
 CV = 47

①  $\frac{150}{5} = 30$

|    | $x$ | $x - \text{mean}$ | $(x - \bar{x})^2$ |
|----|-----|-------------------|-------------------|
|    |     | -20               | 400               |
| 10 | 30  | -10               | 100               |
| 20 | 30  | 0                 | 0                 |
| 30 | 30  | 10                | 100               |
| 40 | 30  | 20                | 400               |
| 50 | 30  |                   |                   |
|    |     |                   | 1000              |

② Variance

$$\frac{1000}{5} = \boxed{200}$$

③ Std. Dev.

$$\sigma = \sqrt{\frac{(x - \bar{x})^2}{n}} = \sqrt{200} = 14.14$$

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$$\textcircled{4} \text{ CV} = \frac{\text{S.D}}{\text{mean}} \times 100$$

$$\frac{14.14}{30} \times 100$$

$$0.47 \times 100 = \boxed{47}$$